Preconditioning kit install guide

Electroniq Buttons Boutique*

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This guide walks you through the steps needed to install the head unit preconditioning kit in your 2021-2024 Ioniq 5. See also the video guides (https://youtube.com/playlist?list=PLYCTmVJH4UYqnw87UZAaqMo1GrOv0cJaL):

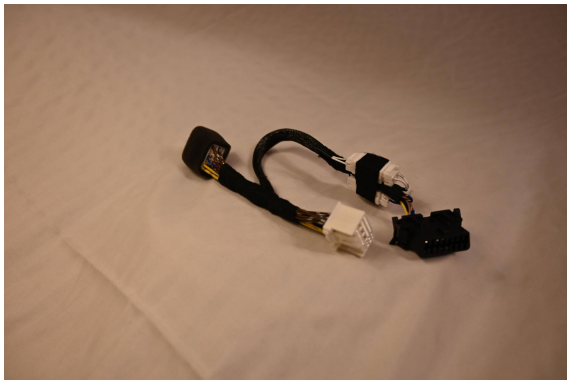
Step 1: Disconnect 12V: video guide: https://youtu.be/ePBdyeBYKk4

Step 2: Trim: Option A: long and careful trim video guide: https://youtu.be/nE4CqGsin7Q

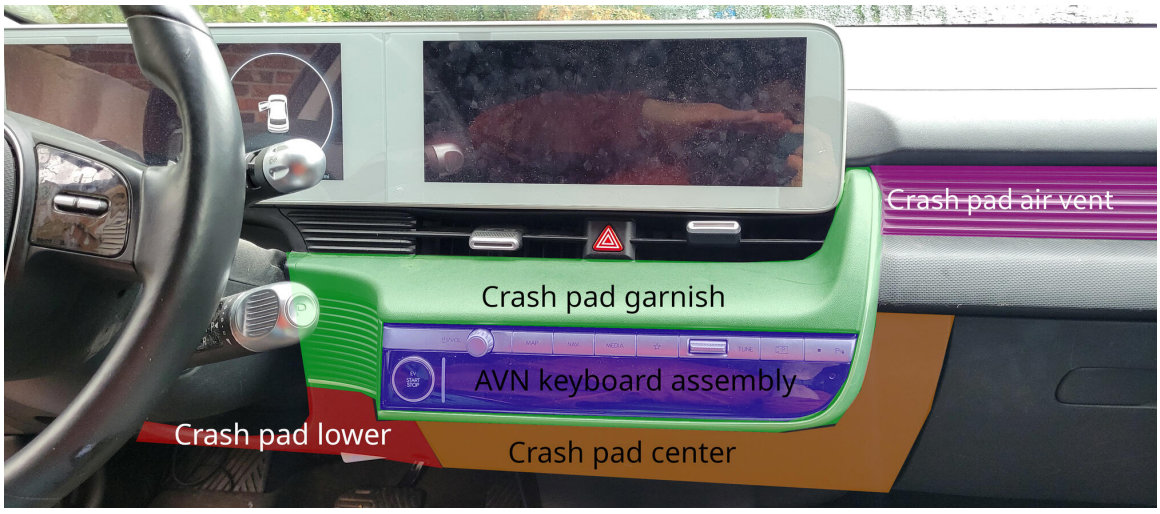
Step 2: Trim: Option B: quick and dirty trim video guide: https://youtu.be/291boryx-Vc

Step 3: Harness Connection: video guide: https://youtu.be/JgrhReAnDGc

Step 4: OBD Mount: video guide: https://youtu.be/PqwYul43jFw



(a)



(b)

Figure 1. (a) Man-in-the-middle harness. (b) Overview of trim panels involved.

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I. INTRODUCTION

The purpose of this guide is to illustrate step-by-step instructions for installing the head unit preconditioning kit, or just the head unit man-in-the-middle (MITM) harness, shown in Figure 1a. In section III, we present three options for trim removal to access the head unit. In section IV, we describe the process of plugging in the harness. In section V, we show attachment of the OBD port and how to plug in the microcontroller.

Necessary tools:

1. 10mm socket or adjustable wrench for 12V battery (not included)
2. Trim removal tools (included if selected)
3. Phillips #2 allen key (included if selected) or similar

Kit components:

1. Head unit harness
2. WiCAN
3. 70mm OBD mount bracket
4. 70mm double-sided tape (two pieces included for redundancy)
5. 4 zip ties

II. STEP 1: REMOVE 12V NEGATIVE TERMINAL

First, always disconnect the 12V battery before removing any electrical connectors! Removing a live electrical connector could cause a discharge and destroy a connector or ECU in your car.

Step-by-step instructions:

1. Open hood (Figure 2a)
2. Open frunk
3. Remove panel covering 12V (Figure 2b)
4. Loosen nut on negative side with a 10mm socket (Figure 2c)
5. Remove negative terminal (Figure 2d)

III. STEP 2: TRIM

The goal here is to get to the treasure chest: the head unit. The proper way involves removal of 4 trim pieces and the AVN keyboard (climate button bar). It's also possible to take a shortcut. The shortcut is much faster but more awkward, with a higher risk of catching a hand on a sharp trim edge. Choose the method that works best for you.

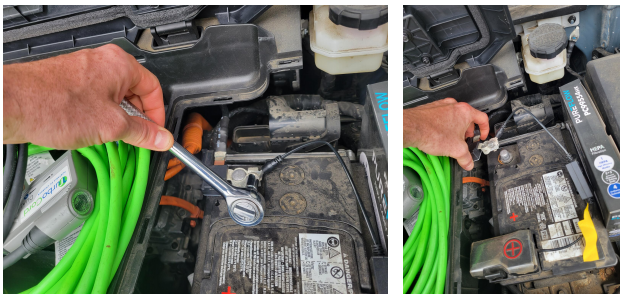
A. Option A: long and careful

Trim panel order of operations:

1. Crash pad lower (Figure 4a)
2. Crash pad air vent (Figure 4b)
3. Crash pad garnish (Figure 4c)
4. AVN keyboard (Figure 4c)



(a) (b)



(c) (d)

Figure 2. (a) Open the hood. (b) Remove panel covering 12V. (c) Loosen nut on negative terminal. (d) Remove negative terminal.



(a)

Figure 3. X marks the head unit.

5. Crash pad center (Figure 4c)

Steps 1 and 2 can be done in either order; both must be done to free the crash pad garnish.
Step-by-step instructions:

1. Remove screw in crash pad lower (Figure 5a)
2. Pull off right half of crash pad lower using trim removal tools
3. While here, remove screw(s) behind crash pad lower (may be 1 or 2; see Figure 5b); the leftmost screw attached



(a) (b) (c)

Figure 4. (a) Crash pad lower. (b) Crash pad air vent. (c) Crash pad garnish, AVN keyboard, and crash pad center.

to the vent behind tends to remain captive, so you have to pull to confirm it's out (Figure 5c)

4. Remove crash pad air vent
5. Remove screw that was hidden behind crash pad air vent (Figure 5d)
6. Remove crash pad garnish
7. Detach AVN keyboard
   (a) Remove left screw (Figure 5e)
   (b) Remove right screw (Figure 5f)
   (c) Remove lower left screw (Figure 5g)
   (d) Remove lower right screw (Figure 5h)
   (e) (optional, check for no 12V power) Remove all electrical connectors from rear of AVN keyboard
   (f) Disconnect trim clips by pulling outwards on the keyboard (Figure 5i)
   (g) If electrical connectors are still connected, avoid putting too much tension on them
8. Remove crash pad center upper right screw that was blocked by AVN keyboard (Figure 5j)
9. (optional) Re-insert AVN keyboard for safekeeping if electrical connectors are still attached
10. Remove crash pad center upper left screw (Figure 5k)
11. Remove crash pad center trim panel using trim removal tools
12. (check for no 12V power first) Disconnect electrical connector from back of panel and move it away from work area

B. Option B: quick

We only partially remove two trim panels with this approach:

1. Crash pad lower
2. Crash pad center

Note that crash pad garnish may still come loose during installation. It can be popped back on when done.
Step-by-step instructions:

1. Remove screw in crash pad lower (Figure 5a)
2. Pull off right half of crash pad lower using trim removal tool
3. While here, remove screw(s) behind crash pad lower (may be 1 or 2; see Figure 5b); the leftmost screw attached to the vent behind tends to remain captive, so you have to pull to confirm it's out (Figure 5c)
4. Pull out the bottom of crash pad center
5. You can now access the head unit from underneath crash pad center



Figure 5. (a) Removing screw in crash pad lower. (b) You may find one or two screws behind crash pad lower. (c) Removing screw(s) behind crash pad lower. (d) Removing screw holding in crash pad garnish. (e) Removing AVN keyboard left screw. (f) Removing AVN keyboard right screw. (g) Removing AVN keyboard lower left screw. (h) Removing AVN keyboard lower right screw. (i) Removing AVN keyboard. (j) Removing crash pad center screw that was blocked by AVN keyboard. (k) Removing crash pad center screw that was behind crash pad garnish, left of the AVN keyboard.

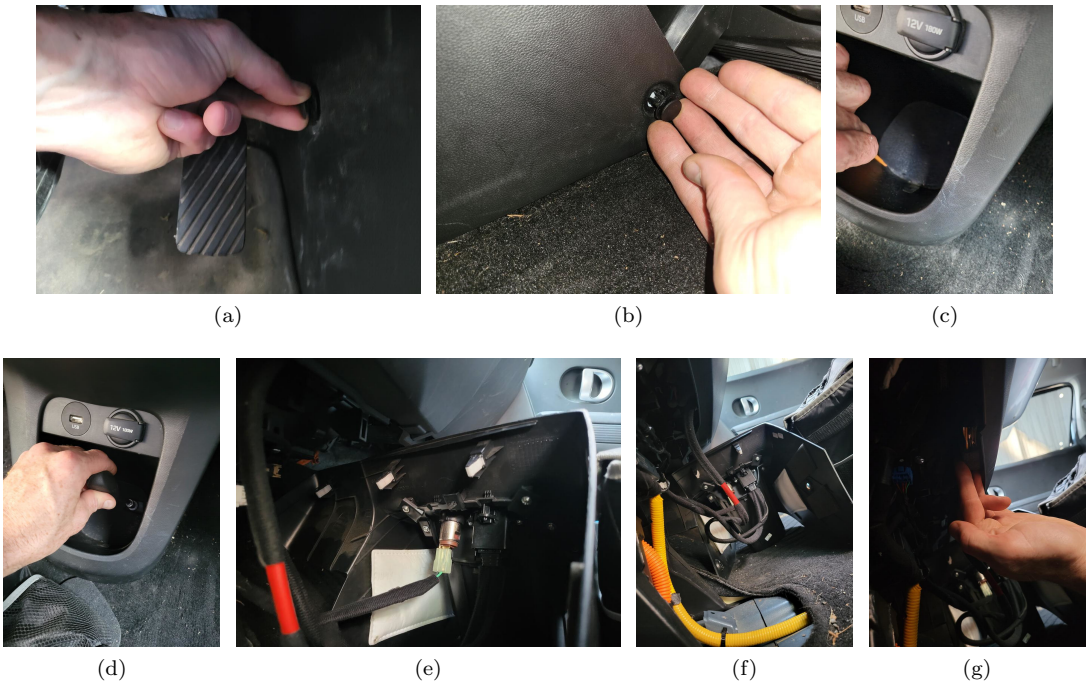
C. Option C: quick option 2

This method isn't recommended, but presented here for information. Reasons to avoid:

1. Requires long straight driver for 10mm socket
2. Doesn't offer access as good as Option B (but could be combined with Option B)
3. Reinstallation of trim is trickier; alignment of the 3 clips at the top is challenging.

The primary value in this method is that well-hidden mount points are possible. These are not documented here, but the EV6 installation manual (https://github.com/tylerharvey/Ioniq5_CAN/blob/main/guides/cars/E-GMP_gen1/EV6/head_unit_preconditioning_kit_EV6_install.pdf) offers some ideas. In this method, we remove crash pad center lower tray, the housing for the front USB port.

1. Remove left trim clip with trim removal tool (Figure 6a)
2. Remove right trim clip with trim removal tool (Figure 6b)
3. Remove floor of console storage with trim removal tool (Figure 6c)
4. Remove bolt with 10mm socket on a long straight driver (Figure 6d)
5. Pry top of crash pad center lower tray to disconnect the 3 trim clips at the top (Figure 6e)
6. Set it to the side (Figure 6f)



(a) (b) (c)

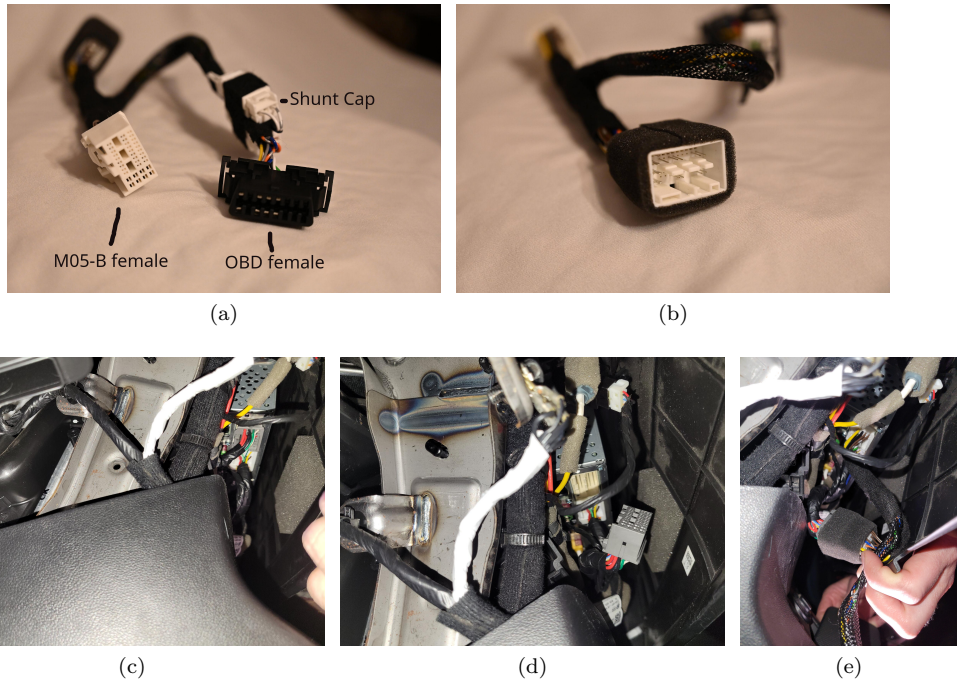
(d) (e) (f) (g)

Figure 6. Removing crash pad center lower tray: (a) Left clip. (b) Right clip. (c) Floor removal. (d) Remove bolt with 10mm socket. (e) Disconnect 3 trim clips with trim removal tool. (f) Place trim to side. (g) You can now reach the bottom of the head unit.

IV. STEP 3: HARNESS CONNECTION

This step covers the installation of the harness into the head unit.
Step-by-step instructions:

1. Identify original grey female head unit connector (Figure 7c)
2. Remove it by squeezing the clip and pulling down (Figure 7d)
3. Insert the harness female side into the head unit (Figure 7a)
4. Insert the female connector you removed into the harness male side (Figure 7e)



(a) (b) (c) (d) (e)

Figure 7. (a) Labeled photo showing female harness connector. (b) Male harness connector. (c) Head unit with connectors plugged in. (d) Female M05-B connector (grey) removed from head unit. (e) Harness installed in head unit.

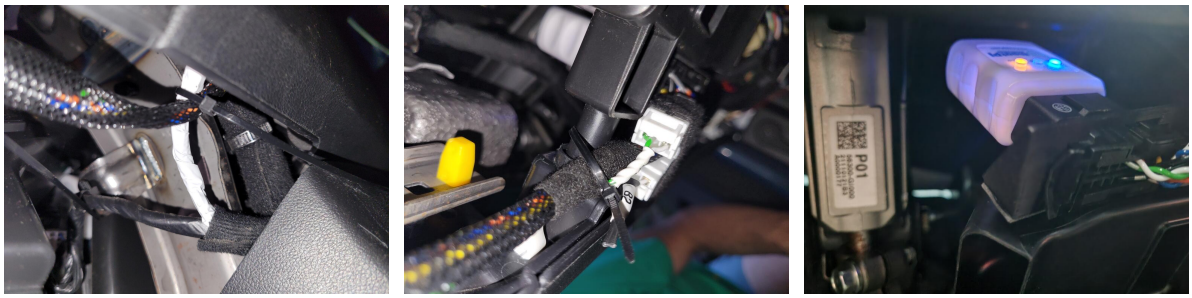
V. STEP 4: OBD MOUNT

In this step, we mount the OBD end of the harness securely and plug in the WiCAN.
Step-by-step instructions:

1. Thread harness along the bottom edge of crash pad lower
2. Plug OBD female into bracket
3. Use double-sided tape to mount bracket to side of footwell vent behind crash pad lower (Figure 8c)
4. Use zip ties to attach harness to other wires and mount points (Figure 8a and 8b)
5. While holding bracket, plug in WiCAN (Figure 8c)

VI. STEP 5: CLEAN UP

Reverse the trim removal steps. Make sure trim clips are aligned, and then firmly pop the trim panels back into place. Reconnect 12V battery negative terminal. The WiCAN should immediately show a blue light.



(a) (b) (c)

Figure 8. (a) Harness zip-tied to existing wire. (b) Harness zip-tied to vent mount. (c) WiCAN mounted on the side of the vent.

A. WiCAN setup

The WiCAN by default provides a WiFi access point. Look for a network named `WiCAN_xxxxxxx`, where `xxxxxxx` is the device ID of your unit, and connect to it with the default password `@meatpi#`. From a phone, the network can take a few seconds to appear, and your phone will warn you that it has no internet access; stay connected anyway. Then open http://192.168.80.1 in a browser to reach the configuration interface. (A WiCAN Pro, not officially supported, uses http://192.168.0.10 instead.)

In the "Settings" tab, change the access point password. With the default password, anyone within WiFi range of your car who knows the default password could unlock your car (yes, this feature is coming!). If you leave the WiCAN in access point mode, we also suggest disabling BLE for now.

Optionally, the WiCAN can be configured in station mode to join your home WiFi so that you can reach it from inside the house. In the "Settings" tab, set "Mode" to "AP+Station", enter your network's "SSID" and "Password", and press "Submit changes". The WiCAN keeps broadcasting its own access point, so you cannot lock yourself out this way. However, access point mode in "AP+Station" can be a bit finicky, so we recommend AP only unless you're using Home Assistant or similar to grab data from your car. The "Status" tab then shows the "Station IP" your router assigned, which is the address to use from your home network.

The full WiCAN WiFi documentation is at https://meatpihq.github.io/wican-fw/config/wifi.